

To Identify Built-ins, Create Dispatch Table, Execute In-Process Commands, Handle Invalid Commands, Maintain State, Test Dispatch Logic

Code:

```
geya@LenovoL480: ~/OS/Skil × + v
GNU nano 8.7.1
#include <stdio.h>
#include <string.h>
#include <unistd.h>
#include <stdlib.h>

void pwd_command() {
    char path[100];

    getcwd(path, sizeof(path));
    printf("Current directory: %s\n", path);
}

void cd_command(char *path) {
    if (chdir(path) == 0)
        printf("Directory changed.\n");
    else
        printf("Invalid directory.\n");
}

void exit_command() {
    printf("Exiting shell...\n");
    exit(0);
}

/* Dispatch table */
struct Command {
    char name[20];
    void (*function)(char *);
};

void execute_command(char *input) {
    char command[20];
```

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```
char argument[80];

sscanf(input, "%s %s", command, argument);

struct Command commands[] = {
    {"pwd", (void *)pwd_command},
    {"cd", cd_command},
    {"exit", (void *)exit_command}
};

int found = 0;

for (int i = 0; i < 3; i++) {
    if (strcmp(command, commands[i].name) == 0) {
        found = 1;

        if (strcmp(command, "cd") == 0)
            commands[i].function(argument);
        else
            commands[i].function(NULL);

        break;
    }
}

if (found == 0)
    printf("Invalid command.\n");
}

int main() {
```

```
int main() {
    char input[100];

    while (1) {
        printf("myshell> ");

        fgets(input, sizeof(input), stdin);

        input[strcspn(input, "\n")] = '\0';

        execute_command(input);
    }

    return 0;
}
```

Output:

```
geya@LenovoL480:~/OS/Skill/Week 5$ ./task2
myshell> pwd
Current directory: /home/geya/OS/Skill/Week 5
myshell> cd ..
Directory changed.
myshell> pwd
Current directory: /home/geya/OS/Skill
myshell> hi
Invalid command.
myshell> exit
Exiting shell...
geya@LenovoL480:~/OS/Skill/Week 5$ |
```